

Reconstructing $\Lambda_b \rightarrow \Lambda \gamma$ at FCC-ee

Stage 1 v3, XGBoost and three-flavour peak optimization

FCC-ee $\Lambda_b \rightarrow \Lambda \gamma$ analysis

4 October 2026

Physics goal and analysis chain

- Reconstruct $\Lambda_b \rightarrow \Lambda(p\pi)\gamma$ and eventually fit the selected $\cos\theta_p$ distribution with a generated-to-reconstructed efficiency and response.
- Chain: named EvtGen decay $\rightarrow$ Delphes IDEA response $\rightarrow$ Stage 1 candidate building $\rightarrow$ Stage 2 offline preselection $\rightarrow$ Zbb-trained XGBoost $\rightarrow$ reconstructed misidentification and photon-pair checks.
- Project the 5.4–5.9 GeV reconstructed signal peak at 6×10^{12} Z decays. Keep signal, wrong combinations, inclusive flavours and forced physics modes distinct.
- All charge conjugates are included. MC identity is attached *after* candidate construction, and is used for diagnostics and signal counting only.

Frozen v3 samples and denominators

Sample	Input events	Generated direct decays	Role
Physics $\Lambda_b \rightarrow \Lambda \gamma$	100,000	100,002	training signal / yield
PHSP $\Lambda_b \rightarrow \Lambda \gamma$	100,000	100,007	angular acceptance
Inclusive Zbb	395,623,929	—	1,091 validated chunks
Inclusive Zcc	499,786,495	—	1,200 validated chunks
Inclusive Zss	499,842,440	—	1,200 validated chunks
Forced $\Lambda \eta$	100,000	100,000	dedicated physics response
Forced $\Lambda \pi^0$	100,000	100,005	dedicated physics response

v3 denotes the reconstruction scenario, not the Stage 0 EDM4hep generator label. Candidate rows and candidate-bearing events have separate counters. Forced modes cannot be added blindly to inclusive Zbb. **A primary Zdd sample has not been processed.**

Stage 1: candidate construction from reconstructed objects

- 1 Choose the primary vertex and reconstructed photon; form opposite-charge track pairs, fit a displaced Λ vertex and test both $p\pi$ mass assignments.
- 2 Retain a good SV ($\chi^2 \leq 9$), daughter $|d_0|/\sigma \geq 3$, PV–SV flight ≥ 0.3 mm and significance ≥ 2 , $|m(p\pi) - m_\Lambda| \leq 15$ MeV.
- 3 Combine the Λ and photon in the same thrust hemisphere, $4.5 \leq m(\Lambda\gamma) \leq 6.5$ GeV; write all candidate observables and keys.
- 4 Stage 2 keeps $|m(p\pi) - m_\Lambda| \leq 12.5$ MeV, $E_{\Lambda_b} \geq 10.5$ GeV and $4.7 \leq m(\Lambda\gamma) \leq 6.5$ GeV. No photon veto enters this offline baseline.

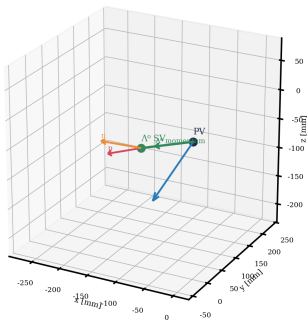
One candidate per downstream row: event entry, candidate slot and candidates in event. Unselected rows are kept for cut studies.

What is measured around one candidate?

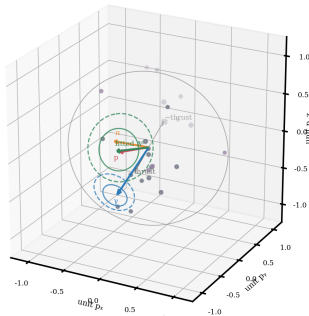
v3 reconstructed $\Lambda b \rightarrow \Lambda^0 \gamma$ candidate • tuple key 4438, raw entry 38885, slot 0
 $m(\Lambda \gamma) = 5.648$ GeV; same thrust hemisphere = True

Fitted vertex topology • PV→SV = 202.91 mm

Momentum directions and isolation



PV/SV positions and flight line are reconstructed.
 Arrows show momentum directions as schematic rays; lengths are scaled.



Blue/green rings: exact ΔR boundaries around $p(\Lambda^*)$ in η - ϕ
 Solid/dashed: $\Delta R = 0.3, 0.5$ • grey ring: thrust hemisphere boundary
 Grey/purple: other charged/neutral particles; crosses: opposite hemisphere
 Arrow lengths normalized; fitted daughter directions shown

Schematic event display: PV, displaced Λ SV, proton/pion trajectories, candidate photon and isolation cones. Geometry illustrates definitions; it is not a detector event.

Stage 1 observable families and construction

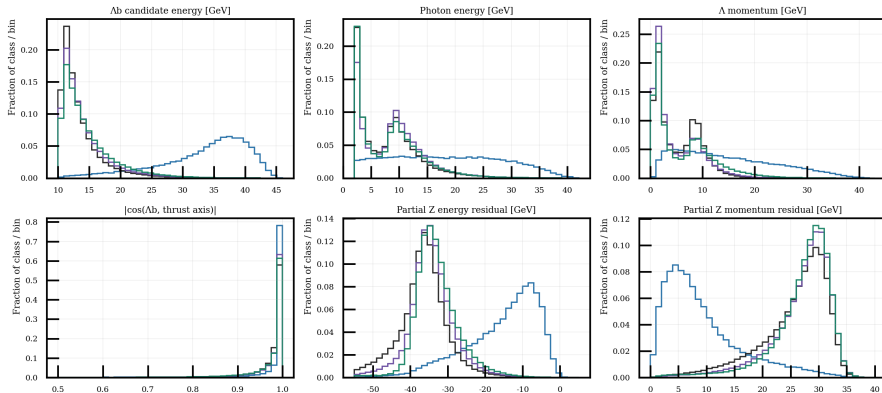
Family	Reconstructed definition / physical role
Kinematics	$p_p, p_\pi, p_\Lambda, E_\gamma, E_{\Lambda_b}$; partial recoil energy/momentum imbalance relative to event hemispheres.
Vertex and topology	Daughter d_0/σ , PV–SV 3D flight significance; thrust-axis alignment of Λ_b .
Isolation and activity	Energy/counts in $\Delta R = 0.2, 0.3, 0.5, 0.7, 1, 2$ around photon/ Λ ; exclude candidate daughters; include displaced neighbour-track counts.
Photon partner	Pair candidate photon with raw same-hemisphere photon; store closest π^0 and η mass distances and minimum $m_{\gamma\gamma}$. Empty partner lists are retained.
Pair identity	Armenteros $\alpha = (p_L^+ - p_L^-)/(p_L^+ + p_L^-)$ and q_T ; computed from reconstructed daughter momenta.

MC parent, grandparent and earlier ancestor fields are appended after reconstruction; none defines candidate or BDT inputs.

Before BDT: kinematic variables in the peak (unit shapes)

v3 candidate shapes before BDT • 5.4–5.9 GeV peak

— Direct $\Lambda_b \rightarrow \Lambda \gamma$ (53,843) — $Z \rightarrow b\bar{b}$ (297,110) — $Z \rightarrow c\bar{c}$ (489,184) — $Z \rightarrow s\bar{s}$ (578,739)

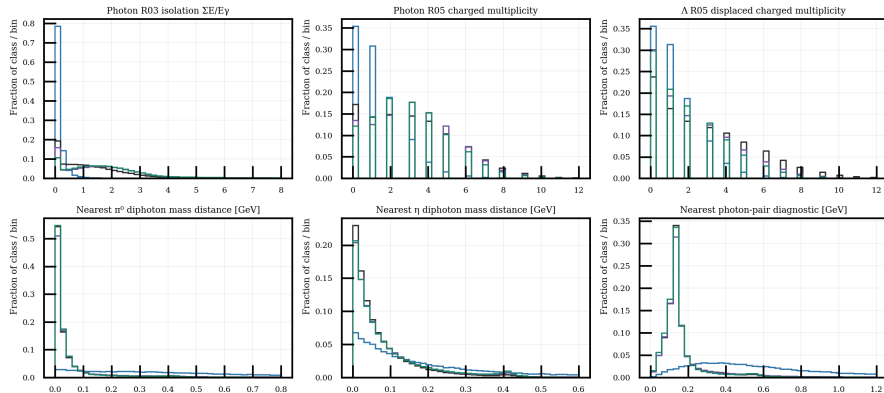


Each class normalized separately; overflow and missing values remain in denominator

Before BDT: isolation and activity in the peak (unit shapes)

v3 candidate shapes before BDT • 5.4–5.9 GeV peak

— Direct $\Lambda_b \rightarrow \Lambda \gamma$ (53,843) — $Z \rightarrow b\bar{b}$ (297,110) — $Z \rightarrow c\bar{c}$ (489,184) — $Z \rightarrow s\bar{s}$ (578,739)

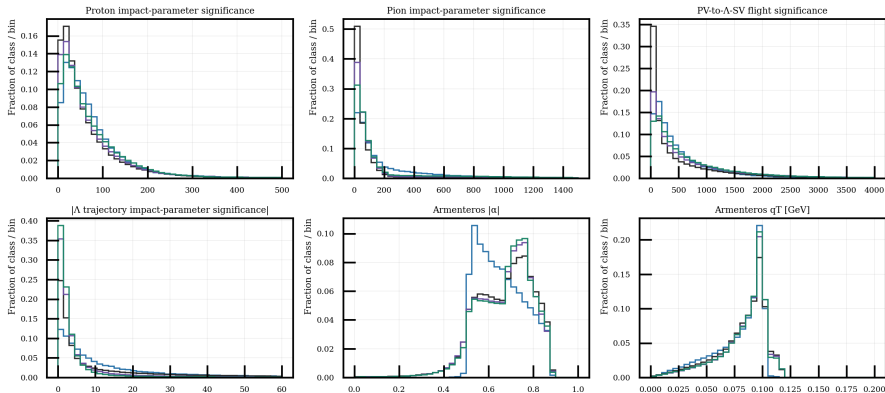


Each class normalized separately; overflow and missing values remain in denominator

Before BDT: displaced topology in the peak (unit shapes)

v3 candidate shapes before BDT • 5.4–5.9 GeV peak

— Direct $\Lambda_b \rightarrow \Lambda \gamma$ (53,843) — $Z \rightarrow b\bar{b}$ (297,110) — $Z \rightarrow c\bar{c}$ (489,184) — $Z \rightarrow s\bar{s}$ (578,739)



Each class normalized separately; overflow and missing values remain in denominator

Frozen XGBoost: 20 reconstructed inputs

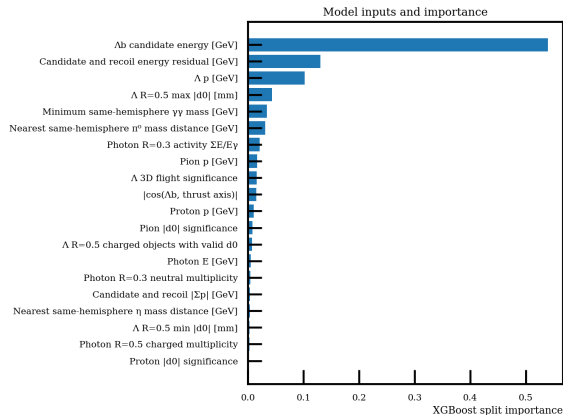
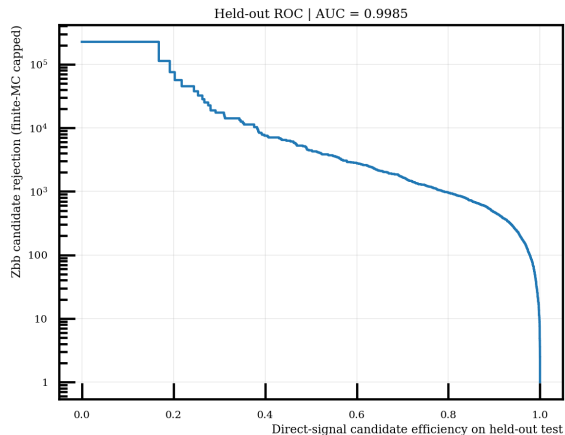
Kinematics and vertex	Isolation, event shape and photon partners
$E_{\Lambda_b}, p_p, p_\pi, E_\gamma, p_\Lambda$	$ \cos\theta(\Lambda_b, \text{thrust}) $
$d_0(p)/\sigma, d_0(\pi)/\sigma$	$E_{\text{all}}^{R<.3}(\gamma)/E_\gamma$
PV–SV flight significance	$N_{\text{charged}}^{R<.5}(\gamma)$
Partial ΔE , partial Δp	$N_{\text{neutral}}^{R<.3}(\gamma)$
	$N_{\text{charged}}^{R<.5}(\Lambda; d_0)$
	min/max $ d_0 $ of Λ neighbours
	$ m_{\gamma\gamma} - m_{\pi^0} , m_{\gamma\gamma} - m_\eta , \min m_{\gamma\gamma}$

Config: `v3_offline_no_veto_proposal`. Isolation removes candidate daughters. Photon-partner distances have a no-partner sentinel. Mass $m(\Lambda\gamma)$, $\cos\theta_p$, truth and candidate keys are excluded from training.

Training and held-out checks

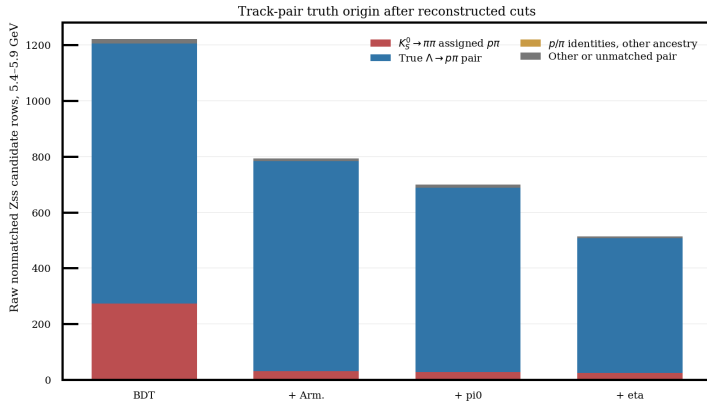
- Train direct Physics truth-matched candidates versus nonmatched inclusive Zbb candidates after the same Stage 2 offline selection. Entire Zbb chunks and signal event hashes stay within one partition.
- The frozen model used 1,091 Zbb chunks, 8 parallel Parquet readers and 16 XGBoost threads. ROC AUC: train 0.998735, validation 0.998617, test 0.998519.
- Model score, feature order, plots, preparation catalog and uncut scored shards are frozen. The 0.953827 score came from the earlier Zbb-only peak scan.
- Current question: can a different threshold improve the *same model* once Zcc and Zss are projected too? A new weighted-mixture model is a separate next study.

Frozen model checks



Left: held-out Zbb-only classification ROC. Right: frozen model feature importance; these are model diagnostics, not a three-flavour performance estimate.

Armenteros: testing $K_S^0 \rightarrow \pi\pi$ misidentification



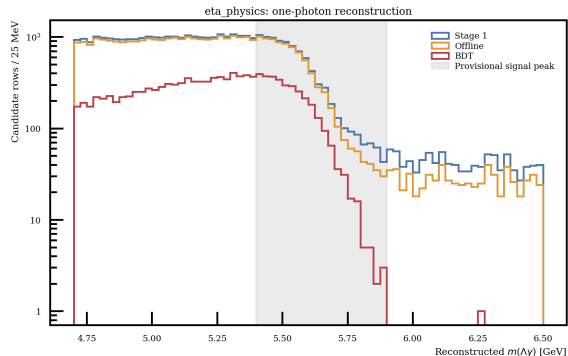
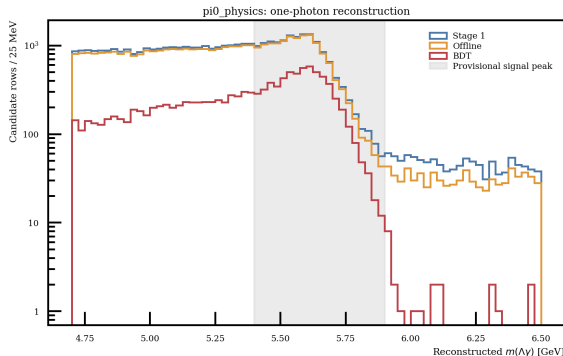
Fixed BDT 0.953827, Z_{ss} peak: 273 true K_S^0 pairs
 $\rightarrow 31$ after rejecting $0.67 \leq |\alpha| \leq 0.78$, $0.075 \leq q_T \leq 0.120$ GeV (88.6% removed). True Λ pairs: 932 $\rightarrow$ 752. Signal peak: 36,900
 $\rightarrow 28,345$ raw direct candidates. Truth categories diagnose the reconstructed cut.

Photon-pair veto mechanics and observed effects

- Search additional raw type-22 photons on the candidate Λ thrust hemisphere. Reject if the nearest reconstructed $m(\gamma\gamma)$ lies within 20 MeV of m_{π^0} or 50 MeV of m_η . No eligible partner means no veto.
- At old BDT+Armenteros, sequential π^0 veto: expected signal 231,473 $\rightarrow$ 228,827 and Zss 1,486,837 $\rightarrow$ 1,308,940 in the peak. Then η veto: signal $\rightarrow$ 194,757; Zss $\rightarrow$ 962,511.
- Forced $\Lambda\eta$ estimate: 9,844 $\rightarrow$ 9,785 $\rightarrow$ 1,924; forced $\Lambda\pi^0$: 84 $\rightarrow$ 66 $\rightarrow$ 60. These are separately normalized sensitivity projections.

These windows are named proposals, not part of Stage 1 or the BDT score. The η step removes sizeable direct signal.

Forced $\Lambda\pi^0$ and $\Lambda\eta$ response (100k each)



Reconstructed candidate mass cutflows, one-photon Stage 1 v3 scenario. Forced samples test veto response; their projected yields are not additional inclusive Zbb components.

Why Z_{cc} and Z_{ss} matter

Old 0.953827 score, peak	Z_{bb}	Z_{cc}	Z_{ss}
Raw rows after BDT	—	469	1,221
Expected after BDT	550,523	677,338	2,286,433
Expected after Armenteros	79,621	410,158	1,486,837
Expected after both photon vetoes	61,422	363,943	962,511

- Scale each flavour with its own processed-event denominator and $\mathcal{B}(Z \rightarrow q\bar{q})$. For Z_{ss} , 0.156 is an assumed per-flavour value from the down-type average; it does *not* include Z_{dd} candidates.
- After all vetoes, 483/514 Z_{ss} peak rows still have a genuine Λ pair; 352/514 selected photons have an immediate recorded s -quark parent. Armenteros cannot remove this genuine- Λ component.

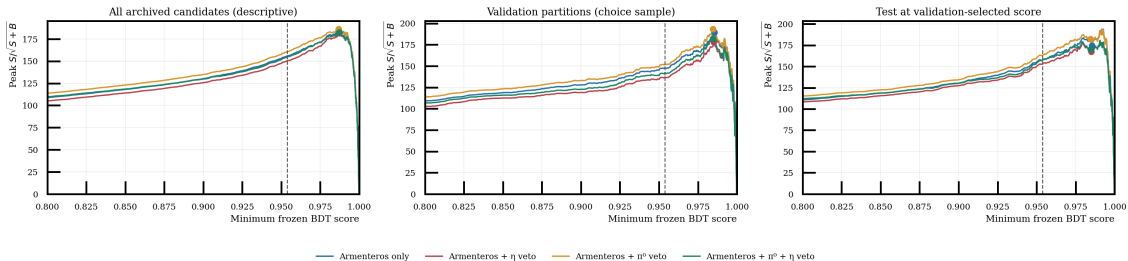
Score scan: the unchanged BDT and three backgrounds

- 1 Keep the Zbb-trained model and score for each scored row. Apply reconstructed Armenteros alone, Armenteros+ η , and π^0 controls on the same rows.
- 2 In the 5.4–5.9 GeV peak, scale direct signal by generated direct decays; scale Zbb, Zcc and Zss separately by $N_Z \mathcal{B}_q / N_{\text{processed},q}$.
- 3 Scan every observed validation score for $S / \sqrt{S + B_{bb} + B_{cc} + B_{ss}}$, then evaluate that fixed score on held-out test partitions. No MC ancestry is an input cut.

$N_Z = 6 \times 10^{12}$, $\mathcal{B}_{bb} = 0.15$, $\mathcal{B}_{cc} = 0.1203$, $\mathcal{B}_{ss} = 0.156$ per-flavour scenario. Zdd has no processed sample and is absent from B . Forced modes and wrong signal combinations are shown separately.

Correct objective: three-flavour $S/\sqrt{S+B}$ versus score

Signal peak 5.4–5.9 GeV • B = Zbb + Zcc + Zss



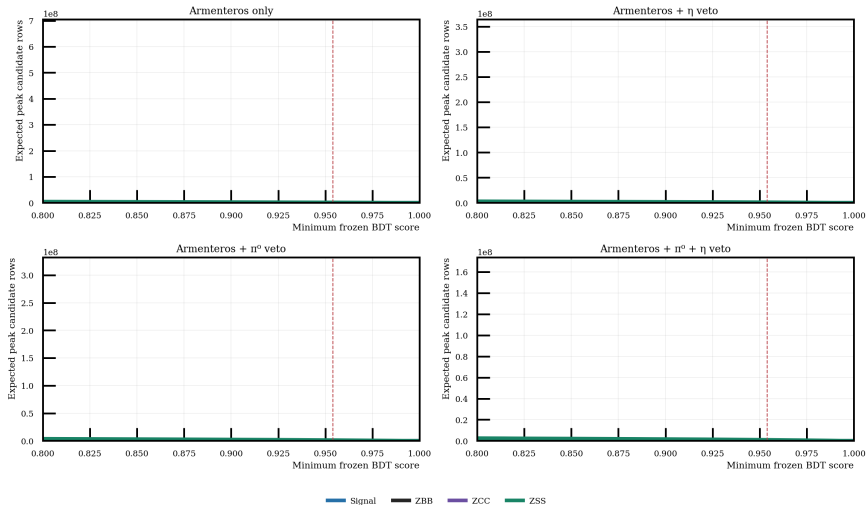
$S/\sqrt{S+B}$ validation choice and held-out test

Selection	Score	Validation FoM	Test FoM	Test $S/(S+B)$
Armenteros, old	.953827	147.8	157.5	10.8%
Armenteros, proposed	.985562	189.3	174.5	20.5%
Armenteros+ η , old	.953827	136.9	153.1	12.0%
Armenteros+ η , proposed	.984990	178.0	167.5	21.0%
Armenteros+ π^0 , exploratory	.984588	193.2	182.4	21.8%

Armenteros-only test FoM improves 10.9% over the old threshold. The η veto lowers the optimized FoM. The π^0 control is promising but needs acceptance and forced-mode review.

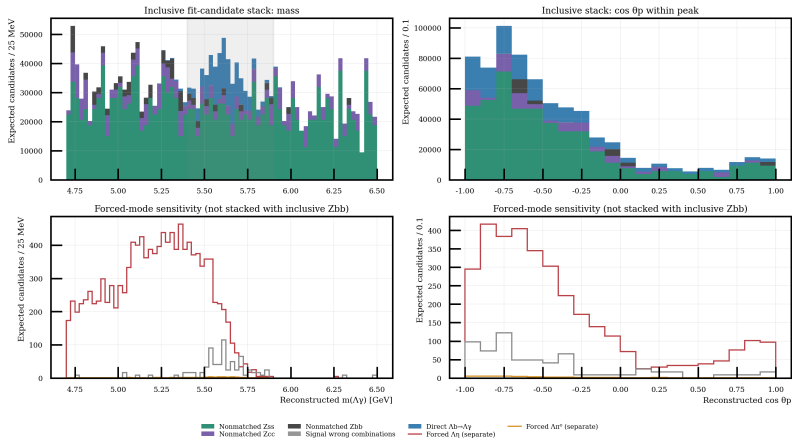
At the Armenteros-only validation optimum there are only 1 Zbb, 5 Zcc, 21 Zss peak rows. Counting, branching and detector uncertainties constrain any claim of an optimum.

Peak signal and each inclusive flavour versus score



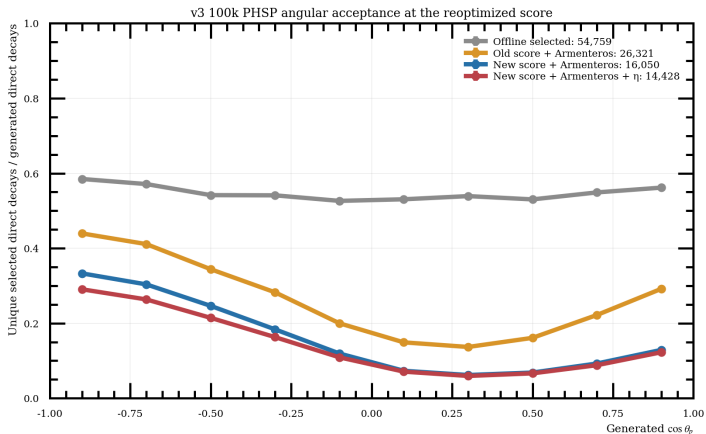
Armenteros-only 0.985562 score: candidate fit view

v3 score ≥ 0.985562 + Armenteros; central expected yields at 6×10^{12} Z



Linear expected candidate yields: direct signal 152,080; Zbb 20,474; Zcc 66,434; Zss 460,657 in 5.4–5.9 GeV. Wrong combinations 629. Forced η 3,347 and π^0 39 are plotted separately and excluded from the inclusive stack.

Armenteros-only 0.985562 score: PHSP acceptance



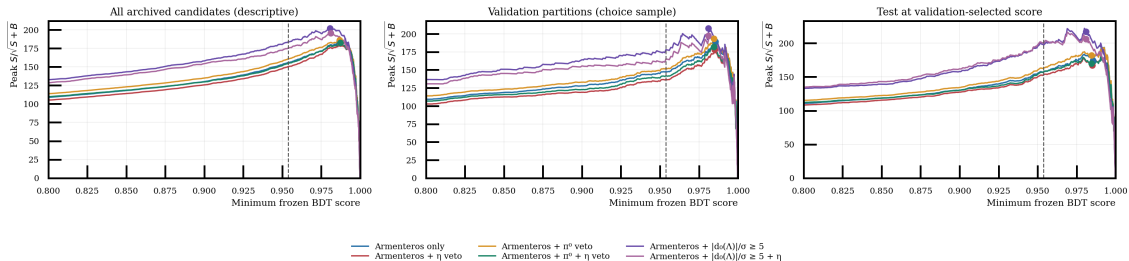
Denominator: 100,007 generated direct PHSP decays before any selection. Old score+Armenteros: 26,321 (26.32%); proposed score+Armenteros: 16,050 (16.05%). The strong $\cos \theta_p$ variation requires binned efficiency in an angular fit.

Alternative offline Λ impact-parameter cut

- The PI's objective is $S/\sqrt{S + B_{bb} + B_{cc} + B_{ss}}$ in the reconstructed 5.4–5.9 GeV peak. The interim $S/\sqrt{B}$ scan and its .991889 point are superseded.
- Compare Armenteros alone with Armenteros plus reconstructed $|d_0(\Lambda)|/\sigma \geq 5$, a threshold from the earlier Zss displacement study. Scan each branch's BDT score on validation; keep the original Zbb-trained model unchanged.
- Λ impact-parameter significance is *not* among the 20 BDT inputs. It is an offline cut on candidate rows after scoring, using no MC identity.

Correct FoM: Armenteros and offline Λ -IP alternatives

Signal peak 5.4–5.9 GeV • B = Zbb + Zcc + Zss



Expected peak yields with the added Λ -IP cut

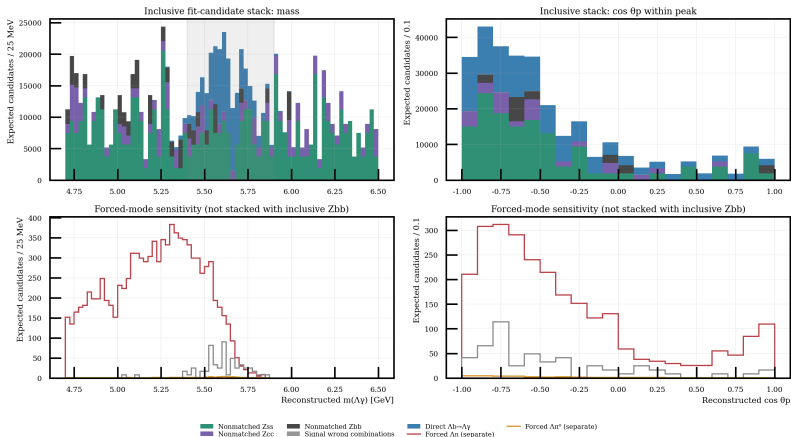
Branch, split	Score	S	Z_{bb}	Z_{cc}	Z_{ss}	$S/\sqrt{S+B}$
+ Arm, validation	.985562	151,496	23,453	71,991	393,348	189.3
+ Arm+IP5, validation	.981096	109,823	0	57,593	112,385	207.6
+ Arm, test	.985562	148,265	11,223	65,086	496,933	174.5
+ Arm+IP5, test	.981096	108,260	11,223	7,232	121,889	217.1

Combined test B : 573,242 without IP and **140,344** with IP; test central FoM rises 24.4% between the separately optimized branches. Full-archive IP5 expectation: signal 110,114; Z_{bb} 18,199; Z_{cc} 30,329; Z_{ss} 140,444. Adding η to IP5 gives test FoM 206.2.

IP5 validation has only 0 Z_{bb} , 4 Z_{cc} , 6 Z_{ss} peak rows; test has 1, 1, 13. The IP threshold was proposed after prior inspection of these campaigns. Its apparent gain requires fresh data and detector-resolution checks.

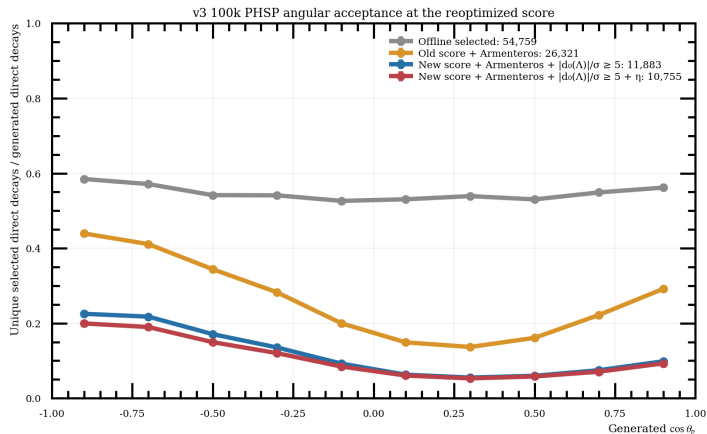
Candidate fit view: Armenteros + offline Λ IP5

v3 score ≥ 0.981096 + Armenteros + $|d_0(\Lambda)|/\sigma \geq 5$; central expected yields at 6×10^{12} Z



.981096 + Armenteros + IP5: signal 110,114; Zbb 18,199; Zcc 30,329; Zss 140,444. Wrong combinations 498. Forced η 2,652 and π^0 34 are separate from the inclusive stack.

PHSP acceptance with the added offline Λ IP5



Score .981096 + Armenteros + IP5 selects 11,883 of 100,007 generated direct PHSP decays (11.88%), versus 16.05% for score .985562 + Armenteros alone. Core 68% $\cos \theta_p$ residual half-width is 0.0115. The fit needs the binned acceptance and response.

Conclusion and PI decision

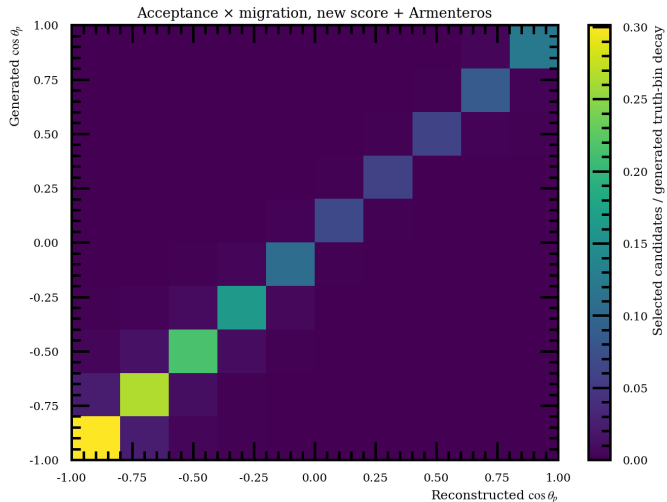
- The corrected peak objective is $S/\sqrt{S + B_{bb} + B_{cc} + B_{ss}}$ after Armenteros. Its no-IP validation score is .985562; the prior .991889 $S/\sqrt{B}$ point is not an optimum for this objective.
- Adding reconstructed $|d_0(\Lambda)|/\sigma \geq 5$ *after* the unchanged BDT gives a validation score .981096 and held-out central FoM 217.1 versus 174.5 without IP. PHSP acceptance falls 16.05%→11.88%.
- The IP result is sparse and post-hoc. Zdd is also missing from B : the Zss sample does not represent it. Require a dedicated Zdd projection, more high-tail MC and PV/SV/track-response checks before adopting a reference cut.
- Compare this offline option with a weighted Zbb/Zcc/Zss BDT and a named Λ -IP feature ablation at matched signal efficiency and angular acceptance.

Backup: exact yield definitions

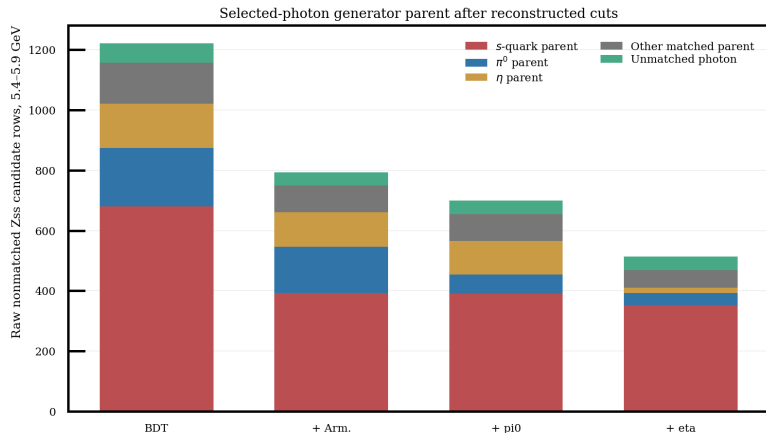
$$w_q = \frac{(6 \times 10^{12})\mathcal{B}(Z \rightarrow q\bar{q})}{N_{\text{processed},q}}, \quad S = (6 \times 10^{12})(0.15)(2)(0.10)(7.1 \times 10^{-6})(0.639) \frac{n_{\text{direct}}}{N_{\text{generated direct}}}.$$

- The signal numerator counts selected direct candidate rows; PHSP acceptance instead counts unique matched generated direct decays. The two denominators answer different questions.
- All yield and FoM values shown are in the 5.4–5.9 GeV reconstructed peak; the stack displays 4.7–6.5 GeV mass and peak $\cos\theta_p$.
- The quoted FoM has candidate-counting uncertainty but omits systematic terms and correlations. The ss branching fraction is an explicit scenario, not a direct measured input.

Backup: PHSP reconstructed versus truth $\cos \theta_p$



Backup: Zss selected-photon ancestor chain



At old BDT+all vetoes, 352/514 photons have immediate recorded s parent; 347 reach Z in four stored ancestor generations. First-parent ancestry supports a strange-parton origin, but is not a full shower graph.

Backup: Delphes IDEA charged-track acceptance

Truth-bin population	Generated	Unique reco match	Fraction
Charged $p_T \geq 0.1$ GeV, $ \eta < 2$	207,781	203,634	98.0%
$2 \leq \eta < 2.56$	5,838	5,382	92.2%
$ \eta \geq 2.56$	2,028	0	0
Fiducial $\Lambda \rightarrow p$	594,874	508,767	85.5%
Fiducial $\Lambda \rightarrow \pi$	577,517	486,799	84.3%
Fiducial $K_S^0 \rightarrow \pi$	575,230	566,990	98.6%

Historical 500k Gamma Delphes response audit; truth-bin matching with unique MCRecoAssociation, not a v3 Stage 1 cutflow. Source: `studies/resolutions/FINDINGS.md`. The measured match fraction includes geometry, reconstruction and association effects.

Backup: displacement and track-resolution checks

- In the same 500k Delphes audit, Λ protons have unique-match fraction 98.7% for production $R_{xy} = 209\text{--}520$ mm, 93.2% for $520\text{--}1,296$ mm, 45.2% for $1,296\text{--}3,226$ mm and 0 beyond $3,226$ mm. Displaced acceptance needs explicit validation for topology cuts.
- Charged tracking card input: $p_T \geq 0.1$ GeV, $|\eta| \leq 2.56$, 2 T field, layered TrackCovariance geometry/material, at least six hits. No single scalar momentum-resolution formula is assumed.
- Resolution audit fits Gaussian cores of $(p_{\text{reco}} - p_{\text{true}})/p_{\text{true}}$ and p_T analogues versus truth p_T , signed η and production radius, using only unique matches. The archived numerical resolution plots are unavailable in this checkout; no widths are inferred from card inputs here.

Source and reproduction scripts: `studies/resolutions/FINDINGS.md`, `run_study.py`, `run-displaced-daughters.py`, `run-comparison.py`.